Theme 1	Analog sensing for field and industrial applications
Theme 2	Digital logic systems for automation and safety control.
Theme 3	Audio-based human interaction and safety sensing systems.
Theme 4	Discrete Logic Design and VLSI Principles for Industrial Systems
Theme 5	Fail-safe timing and power management protection systems.

Theme 1: Analog sensing for industrial applications.

1. DC Line Status Indicator

In many industrial control panels, technicians need a quick visual indication of whether a DC supply line is working correctly. Design a simple circuit that monitors a DC voltage (5–24 V) and indicates its condition using LEDs. The circuit should display LOW, NORMAL, or HIGH voltage levels using resistor dividers and basic comparator or transistor threshold stages so that the status of the line can be identified instantly without using a multimeter.

2. Portable Battery Checker

Technicians often carry spare batteries but cannot easily determine whether they are still usable. Design a compact battery tester that checks the condition of AA or 9 V batteries. The circuit should apply a small load to the battery and use voltage thresholds to indicate GOOD, WEAK, or DEAD using LEDs, allowing users to quickly decide whether the battery should be kept or replaced.

3. Heat Sink Temperature Indicator

In electronics labs and small power circuits, components such as heat sinks can become hot without obvious warning. Design a temperature monitoring circuit using an NTC thermistor that indicates temperature levels through LEDs. The circuit should provide three clear states such as SAFE, WARM, and HOT, helping users monitor temperature visually without using a digital thermometer.

4. Ambient Light Level Indicator

Workspaces such as soldering benches or small indoor setups often require sufficient lighting. Design a circuit using an LDR (Light Dependent Resistor) that measures ambient light and displays the brightness level using LEDs. The circuit should indicate whether the lighting condition is DIM, NORMAL, or BRIGHT, helping users quickly judge if additional lighting is needed.

5. AC Presence Indicator

In electronics labs, it is often necessary to confirm whether an AC signal or supply is present at a test point. Design a circuit that detects the presence of a low-voltage AC signal and lights an LED when AC is detected. The circuit should include basic rectification and filtering so that the LED turns on only when a sufficient AC voltage is present.

6. Simple Water Level Indicator

Water tanks in homes or small industrial setups require simple monitoring to prevent overflow or empty conditions. Design a water level indicator using conductive probes or simple sensors that detect different water levels and display them using LEDs (e.g., LOW, MEDIUM, FULL). The circuit should provide a clear visual indication of the water level in the container.

7. Battery Voltage Monitor

Many portable devices stop working properly when the battery voltage becomes too low. Design a circuit that continuously monitors a battery voltage and lights an LED when the voltage falls below a predefined

threshold. This simple low-battery warning system can help users know when the battery needs to be replaced or recharged.

8. Overheat Warning Indicator

Electronic components such as motors or power transistors may fail if they become too hot. Design a circuit that uses a temperature sensor (such as a thermistor) to trigger a warning LED when the temperature exceeds a safe limit. The circuit should act as a simple overheat alert system.

9. DC Power Supply Indicator

In laboratory setups with multiple power rails, it is useful to know whether a power supply is active. Design a circuit that monitors a DC power supply line and lights an LED whenever the supply voltage is present and above a minimum threshold. This indicator should help users quickly confirm that a power source is operating correctly.

10. Inductive Load Activity Indicator

In many circuits, inductive loads such as motors or relays are used. Design a simple sensing circuit that detects when current is flowing through an inductive load and indicates it using an LED. The circuit should provide a visual indication that the load is active.

Theme 2: Digiatal logic systems for automation and safety control.

1. At a busy airport, multiple aircraft may request permission to land at the same time. The airport has two runways available, allowing two aircraft to land simultaneously. When landing requests are received, the control system must decide which two aircraft should be granted landing clearance based on predefined selection rules. The system analyses the incoming request signals and assigns the two available runways to the appropriate aircraft

while preventing other aircraft from landing at that moment. Design a digital circuit that detects the landing requests and generates the runway assignment signals for the selected aircraft.

2. In a digital system, a controller selects one device from several devices using a binary input. The control signal is sent through a distribution circuit so that only the selected device receives the signal. To check whether the correct device was activated, the active device line is sent to an identification circuit that converts it into a binary code. This generated code is then compared with the original selection input to verify that the signal reached the correct device. Design a digital circuit that performs this activation and verification process.(demux+encoder)

3. Binary Combination Lock with Decoy Inputs
   Standard code locks are vulnerable to trial entry. Design a logic lock where four inputs must match a hardwired pattern, but two additional decoy inputs trigger an alarm latch if touched at all — requiring the user to both know the correct combination and actively avoid the decoys, using only gates and a latch.

4. In asynchronous digital systems, two signals competing to arrive first must be resolved without a clock. Implement a metastability-aware arbiter using cross-coupled gates that determines which of two input edges arrived first and latches the winner, with a brief lockout preventing re-triggering — directly demonstrating a real problem in digital design.

5. In lab experiments and small test jigs, instructors often need to count how many times a button was pressed or a sensor was triggered without using microcontrollers. Design a 0–9 event counter using a counter IC (such as CD4029/7490/CD4017) with a debounced pushbutton input and LED indication, ensuring each count from 0 to 9 has a unique binary or decoded LED pattern and can be reliably reset to zero during demonstrations.

6. Design a basic arithmetic logic unit that accepts two single-bit binary inputs and a carry/borrow input and produces a result based on selected operations. The unit should be able to perform logical operations such as AND and OR, as well as arithmetic operations such as addition and subtraction. The operation performed by the unit should depend on control signals that select which function is executed. The circuit should generate the correct output result along with the carry or borrow signal when performing arithmetic operations. This design demonstrates how fundamental logic and arithmetic functions can be combined to form the core processing element of a simple CPU

7. A digital control system must track the remaining steps of an operation before restarting the process. The system begins from its maximum state and decreases its value at every clock pulse until it reaches zero, after which the sequence repeats automatically.

   Design a digital circuit that generates the following binary sequence:

   111 → 110 → 101 → 100 → 011 → 010 → 001 → 000 → repeat..

8. In a digital communication system, a small data buffer is used to temporarily store a 3-bit data word before transmitting it to another module. Depending on the control signal provided by the controller, the stored data must shift either to the left or to the right during each clock pulse. When shifting left, the bits move toward the higher position, and when shifting right, the bits move toward the lower position while a new bit can enter the register.

   Design a sequential circuit that stores a 3-bit data value and shifts the data either left or right based on a control input. The circuit should update the stored value on every clock pulse and allow observation of the shifted data at the output.

9. In a digital communication system, data may be received from a processing unit as a parallel word but must be transmitted through a communication channel in a serial form. To support this operation, a temporary storage circuit is required that can load multiple bits of data simultaneously and then shift them out one bit at a time during successive clock pulses. Design a sequential circuit that can load a 3-bit data word in parallel and subsequently shift the stored data serially through the register based on a control signal. The circuit should update its contents on each clock pulse and allow observation of the serial output.

Theme 3: Audio-based human interaction and safety sensing systems.

1. Single-Transistor Audio Tester. In audio labs and repair shops, technicians often need to quickly check whether a microphone or small audio source is alive without setting up an amplifier and speaker system. Design a single-transistor audio tester in common-emitter configuration that takes input from a microphone or audio line and drives a small speaker or buzzer, providing a clear audible difference between silence, weak signals, and normal speech or tones.
2. Tap-to-Test Microphone Checker. Stage technicians and students regularly plug in electret microphones but cannot easily confirm if the mic capsule and wiring are intact before an event. Develop a “tap-to-test” microphone checker using an electret microphone, transistor amplifier, rectifier, and hold capacitor that flashes an LED reliably on taps or loud speech while ignoring normal background noise.
3. IR Remote Control Tester. Many households and service centers have multiple IR remote controls but no quick way to verify if a remote is actually transmitting when a button is pressed. Create an IR remote-control tester using a photodiode or phototransistor front end and an NPN amplifier to drive an LED so that whenever a remote is pointed at the tester and a button is pressed, the LED blinks in visible bursts confirming remote functionality.

4. 3-LED Audio Level Indicator. In DIY music practice rooms, musicians often want a simple visual reference to know whether their input signal to an amplifier is too weak, ideal, or clipping. Implement a 3-LED input level indicator using diode rectification and comparator thresholds to indicate “low signal,” “optimal level,” and “clipping risk.”
5. Adjustable Noise Level Warning Indicator. Workshops, labs, and makerspaces need a simple way to indicate when ambient noise exceeds a safe or comfortable level. Build a noise level warning circuit using an electret microphone, amplifier, rectifier, and adjustable threshold transistor stage that turns an LED ON when the sound level crosses the limit.
6. Dual-Stage Audio Amplifier for Weak Sound Detection. In many environments such as small electronics laboratories, workshops, or field repair situations, very weak audio signals from microphones or sensors are difficult to observe or monitor directly. A single amplification stage may not provide sufficient gain to clearly detect these signals. Design a dual-stage audio amplifier that can effectively amplify weak audio inputs so they can be easily monitored through a small speaker or indicator. The system should improve signal strength while maintaining acceptable sound quality, enabling users to clearly distinguish between very weak signals, normal speech, and silence during testing or monitoring applications.
7. Sound-Activated Oscillating Alarm. In safety applications, a loud sound should trigger a repeating alarm rather than a single beep. Create a sound-activated oscillator alarm using a microphone amplifier and envelope detector that enables an astable multivibrator driving a buzzer.
8. Dual-Threshold Audio Indicator. For audio testing setups, users may want to know whether the signal is too weak or too strong. Design a dual-threshold audio indicator using a microphone amplifier and two comparator stages that drive separate LEDs for low and high signal levels.
9. Whistle-Activated Safety Indicator. In environments such as industrial workshops or garages, workers may wear gloves or have dirty hands, making it inconvenient to operate switches. Design a whistle-activated safety indicator that detects a high-frequency

whistle tone using a microphone, frequency-selective filter, amplifier, and threshold detection stage to activate an LED or buzzer.

10. Sound-Triggered Door Knock Indicator. In offices or studios where doors remain closed, knocks may not always be heard clearly. Build a sound-triggered knock detection indicator using a microphone sensor, signal amplifier, and pulse shaping circuit that flashes an LED whenever a strong knock is detected on the door.

Theme 4: Discrete Logic Design and VLSI Principles for Industrial Systems.

Note: (Strictly no IC's Should be used)

1. Smart Device Activation Logic *(Universal Logic Using NAND)*

In many embedded systems, devices must activate only under specific conditions determined by multiple control signals. Design a transistor-level logic circuit that activates an output LED when a sensor signal B is active while safety input A is inactive, or when an override input C is active. The circuit must implement the Boolean function $F(A,B,C) = \overline{A}B + AC$ using only NAND-based transistor structures, demonstrating how universal gates can be used to implement complex logic functions.

2. Industrial Machine Safety Logic *(Logic Function Implementation)*

Industrial machines often require multiple safety conditions to be satisfied before operation is permitted. Design a transistor-level logic circuit that controls a machine indicator LED based on the Boolean expression $F(A,B,C) = AB + \overline{A}C$, where the output becomes active when both safety switch A and control input B are active, or when the safety switch is inactive but an emergency override signal C is present. The circuit should clearly demonstrate the implementation of Boolean expressions using transistor logic.

3. Request Priority Controller *(4-Input Priority Encoder)*

In digital communication systems and microprocessors, multiple subsystems may request access to a shared resource at the same time. Design a 4-input priority encoder using transistor logic that identifies the highest-priority active input among four request signals and generates the corresponding binary output. LED indicators should be used to display the encoded output representing the selected request.

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4. Digital Level Comparator for Monitoring *(2-Bit Binary Comparator)*

In monitoring and control systems, digital values from sensors or counters are often compared to determine relative magnitude. Design a transistor-based digital circuit capable of comparing two 2-bit binary inputs and indicating whether one value is greater than, equal to, or less than the other. Separate LEDs should clearly display each of these comparison results.

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5. Simple Arithmetic Processor Block *(Half Adder)*

Arithmetic operations form the core of digital processors and computation units. Design a half adder using transistor-level logic gates that accepts two binary inputs and produces SUM and CARRY outputs. Both outputs should be indicated using LEDs to demonstrate the basic arithmetic operation performed by digital logic circuits.

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6. Memory Element for Digital Systems *(SR Latch)*

Digital systems often require a basic memory element to store the current state of a signal or control input. Design an SR latch using cross-coupled transistor logic capable of storing a single bit of information. The circuit should maintain its output state until new

input signals modify the stored value, and the stored output should be displayed using LEDs.

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7. Communication Error Detection Circuit *(Parity Generator)*

Digital communication systems frequently use parity bits to detect transmission errors in binary data. Design a 3-bit even parity generator using transistor logic that produces an output parity bit ensuring that the total number of logic HIGH signals in the system remains even. The generated parity output should be displayed using an LED.

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8. Digital Signal Selector *(2-to-1 Multiplexer)*

Electronic systems often require selecting one signal from multiple inputs, such as switching between sensors or communication lines. Design a 2-to-1 multiplexer using transistor logic in which a control signal determines which input is routed to the output. The selected output signal should be indicated using an LED.

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9. Oscillator for Clock Generation *(Ring Oscillator)*

Digital systems require clock signals to synchronize operations such as counting, data transfer, and processing. Design a ring oscillator using an odd number of transistor inverter stages connected in a loop, generating a continuous oscillating signal. The oscillation should be observable using an LED indicator or measurement point.

10. Event Counter for Monitoring *(2-Bit Binary Counter)*

Counters are widely used in digital systems to track events such as button presses, pulses, or sensor activations. Design a 2-bit binary counter using transistor-based sequential logic that

generates the sequence 00 → 01 → 10 → 11 → repeat, with LEDs used to display the current count value.

Theme 5: Fail-safe timing and power management protection systems.

1. Many small DC loads like fans and lamps draw a high inrush current if turned on immediately at power-up, potentially stressing lab supplies or demo setups. Design an ON-delay start module using an NE555 monostable, RC timing network, and transistor/relay or MOSFET that waits for a preset time before energizing the load, so that loads can be sequenced smoothly in lab experiments or small control panels.

2. Equipment such as soldering stations or power amplifiers often need their cooling fans or pumps to continue running for a short time after the main power is switched off to prevent heat build-up. Develop an OFF delay stop module using an NE555 monostable that is triggered when the main control signal goes low and keeps the fan or pump ON for an adjustable interval, providing repeatable cooling behaviour without any microcontroller.

3. Simple DC motor speed control or LED strip dimming in hobby projects is often done by varying voltage, which is inefficient and unstable. Create an adjustable duty-cycle PWM controller using an NE555 in astable mode, with a potentiometer and diodes to separately control charge and discharge times of the timing capacitor, driving a transistor/ MOSFET so that users can smoothly adjust motor speed or light brightness at constant supply voltage

4. In sensor systems with very narrow pulses (e.g., from encoders or limit switches), LEDs or slow circuits cannot reliably show or process such short events. Implement a pulse-stretching circuit using a 555 or transistor monostable that takes very short input pulses and outputs longer, fixed-width pulses that can be easily observed on an LED or interfaced to slower logic, making invisible events visible for debugging.

5. Embedded controllers and PLCs often provide a "heartbeat" signal to show they are running, but basic indicator LEDs cannot reveal when this periodic signal silently stops. Build a heartbeat watchdog circuit for a logic line using RC charge/decay and a transistor or 555, such that an LED remains ON while pulses keep arriving and automatically turns OFF after a designed timeout when pulses stop, giving a clear visual indication of system hang or failure.

6. In battery-powered field equipment (like solar-charged lanterns), lead-acid or Li-ion batteries can be permanently damaged if they are discharged below a critical voltage. Simple voltage dividers don't account for "voltage bounce" when a load is removed, causing the system to flicker on and off. Design a circuit using a voltage comparator (or a 555 used as a Schmitt trigger) and a MOSFET that disconnects the load when the battery hits a "low" threshold. It must include hysteresis, meaning the load stays off until the battery has recharged to a significantly higher, safe level, preventing rapid oscillations.

7. Standard fuses are slow and must be replaced after a fault. In test environments, users often accidentally short-circuit outputs. A "crowbar" circuit is too aggressive, and a simple limiter generates too much heat. Implement a "Hiccup Mode" power protector. If the current exceeds a set limit (sensed via a low-ohm resistor and a transistor), the circuit should immediately cut power and then wait for a 5-second "cool down" period before automatically attempting to restart. If the short is still there, it should cut power again and repeat the cycle.

8. High-power LED strips are often driven by unregulated power supplies. A small spike in voltage can lead to "thermal runaway," where the LED draws more current, gets hotter, and eventually burns out. Resistors are inefficient for this. Design a constant-current limiter using two NPN transistors and a sense resistor. The circuit should "clamp" the current at exactly 100mA, regardless of whether the input voltage is 9V or 15V

9. In industrial presses, a operator must push a button to cycle the machine. If they tape the button down to work faster, it bypasses safety. The machine should only cycle once per press. Design a "Monostable"

circuit that produces a single 500ms pulse regardless of whether the input button is tapped for 1ms or held down for 1 hour. The circuit must "reset" only after the button is physically released.

10. In sensor monitoring, sometimes you need to capture the highest voltage "spike" that occurred while you weren't looking (e.g., a pressure surge). Design a circuit using an Op-Amp, a Diode, and a Capacitor that "holds" the highest voltage it has seen at its input.